

HW3

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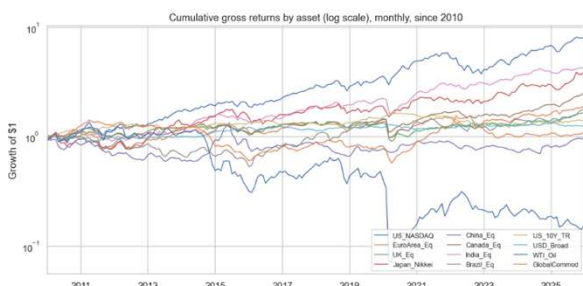
Macro Factors

Asset Universe & Macro 3-Factor Model

- In-sample Jan 2010 – Dec 2024, Out-of-sample Jan 2025 – Jan 2026

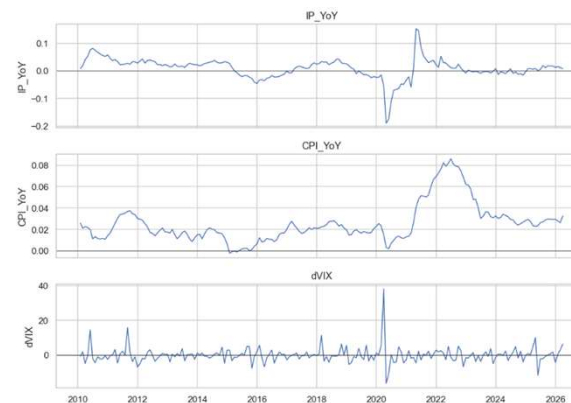
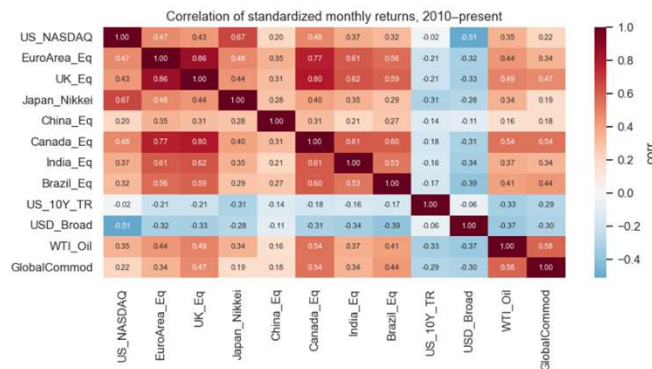
- Take log returns and standardized on:

- Equities: **US, Euro, UK, Japan, China, Canada, India, Brazil**
- Rates: **US 10Y Treasury total return**
- FX: **Broad USD index**
- Commodities: **WTI Crude Oil**



- NASDAQ strongest, Brazil/China weakest
- Stress periods: 2014-2016 Oil collapse, 2020 COVID

- Equities + correlated
- Commodity + to Oil/Equities
- USD - to non-US equities
- Treasuries are defensive



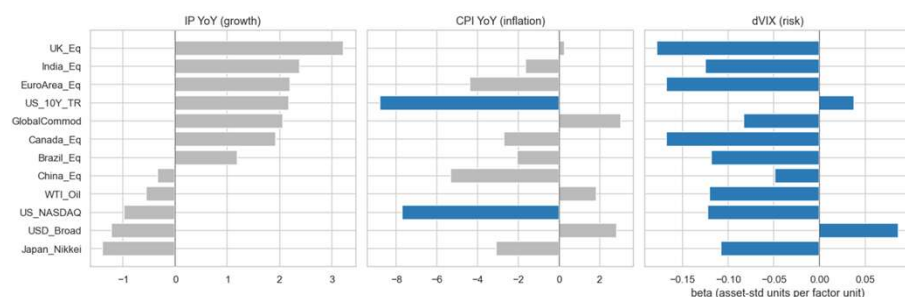
	Growth (Industrial production)	Inflation (CPI)	Risk (VIX)
Expected Loading	+ Equity + Commodity	– Bond + Commodity + USD	– Equity + Bond – Commodity + USD
Historical Signals	2020 contraction, 2021 rebound	2021–22 inflation surge	2018 selloff, COVID, 2022 volatility
What it captures	Business-cycle dynamics	Inflation regime shifts & price pressure	Financial stress

Macro Factors

In-sample VS Out-of-sample Regression

$$\tilde{r}_{i,t} = \alpha_i + \beta_i^{\text{IP}} \Delta \text{IP}_t + \beta_i^{\text{CPI}} \Delta \text{CPI}_t + \beta_i^{\text{VIX}} \Delta \text{VIX}_t + \varepsilon_{i,t}$$

In-sample factor loadings (blue = $|\beta| \geq 2$ with HAC SE)



Growth ★	Inflation ★★	Risk ★★★
- Equity loadings + - Statistical significance weak	10Y Treasury - - Commodity +	- Significant for 10 of 12 assets - Equity loadings - - USD loading +

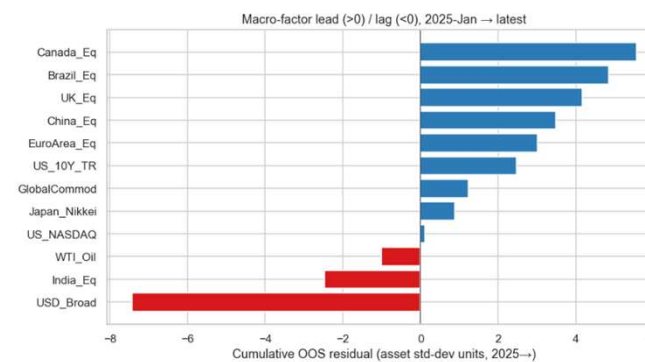
Model Fit (R^2)

- Strongest:** UK Equity, Canada Equity, Euro Area Equity
- Weakest:** US 10Y Treasury, China Equity

Macro factors explain **6%–64%** of monthly variation, but substantial residual variation remains → need PCA analysis

$$\text{Residual}_{i,t} = \text{Actual Return}_{i,t} - \text{Predicted Return}_{i,t}$$

- With sufficient observations and complete business cycles, should **keep the full sample** while treat 2020–2022 cautiously as a structural break

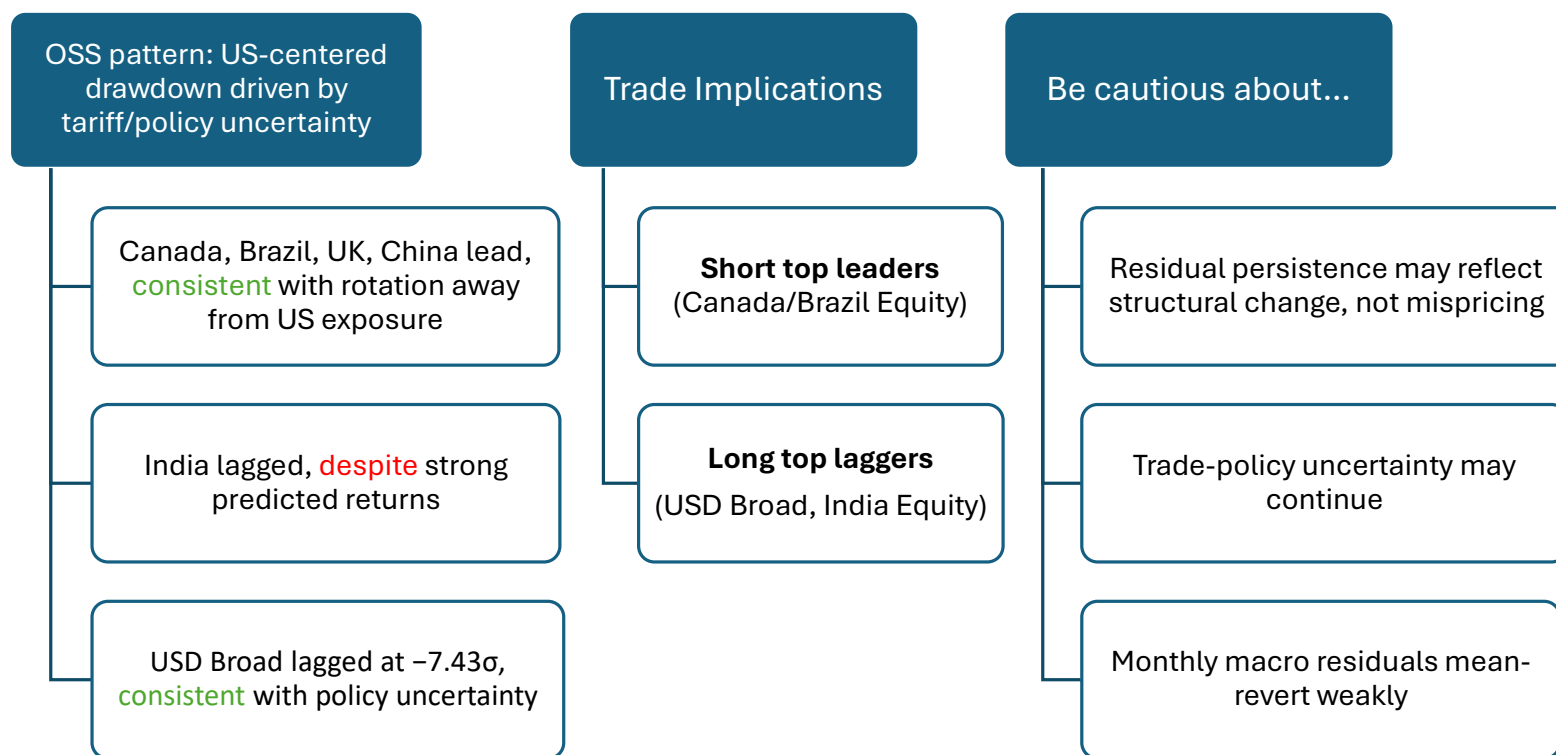


Leaders (Residual > 0)	Neutral (Residual ≈ 0)	Lagers (Residual < 0)
Canada +5.56σ	NASDAQ +0.12σ	USD Broad -7.43σ
Brazil +4.84σ		India -2.46σ
UK +4.16σ		
China +3.48σ		

Interpretation may be limited by the short 12-month OOS window.

Macro Factors

Interpretations & Trade Implications



- The residuals are better interpreted as signals of **omitted** structural risk factors, rather than standalone trading signals.

PCA on the Same In-sample Window

	PC1	PC2	PC3
US_NASDAQ	-0.27	-0.50	0.21
EuroArea_Eq	-0.36	-0.05	0.07
UK_Eq	-0.37	0.04	-0.05
Japan_Nikkei	-0.26	-0.21	0.53
China_Eq	-0.17	0.04	0.46
Canada_Eq	-0.37	0.06	-0.14
India_Eq	-0.32	-0.03	-0.15
Brazil_Eq	-0.31	0.06	-0.21
US_10Y_TR	0.14	-0.59	-0.44
USD_Broad	0.23	0.40	0.25
WTI_Oil	-0.29	0.22	-0.14
GlobalCommod	-0.26	0.37	-0.30

PC 1



Global Risk Factor (45.7%)

- + Equity, commodity
- USD & Treasury

PC 2



USD / Commodity Factor (10.7%)

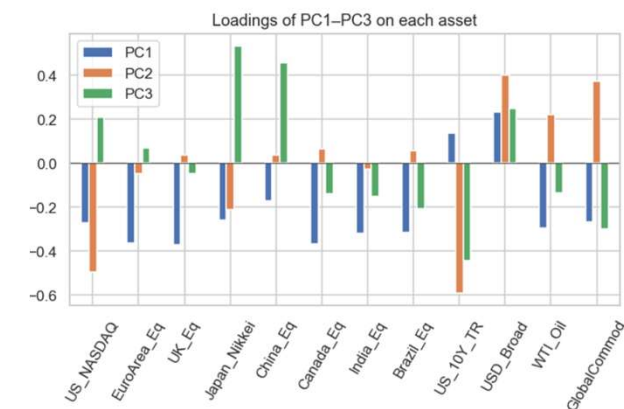
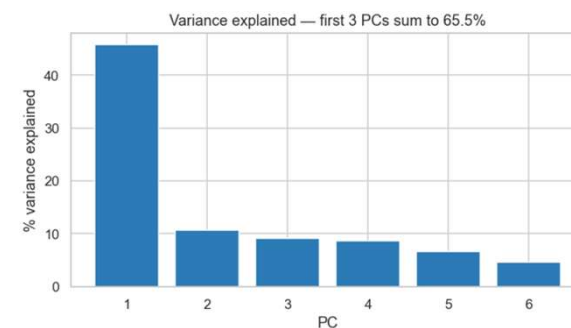
- + USD, Oil, Commodity
- NASDAQ, Japan, Treasury

PC 3



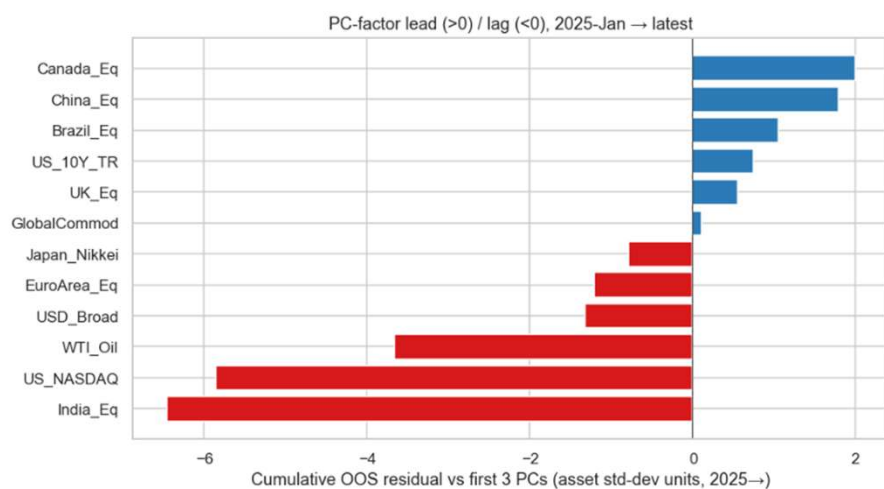
Regional Rotation Factor (9.1%)

- + Japan, China, NASDAQ, USD
- Brazil, India, Canada, Oil, Commodity



PC-Implied OOS Performance & Interpretation

- **Top Leaders:** Canada/China/Brazil Equity
- **Top Laggards:** India Equity, NASDAQ, Oil



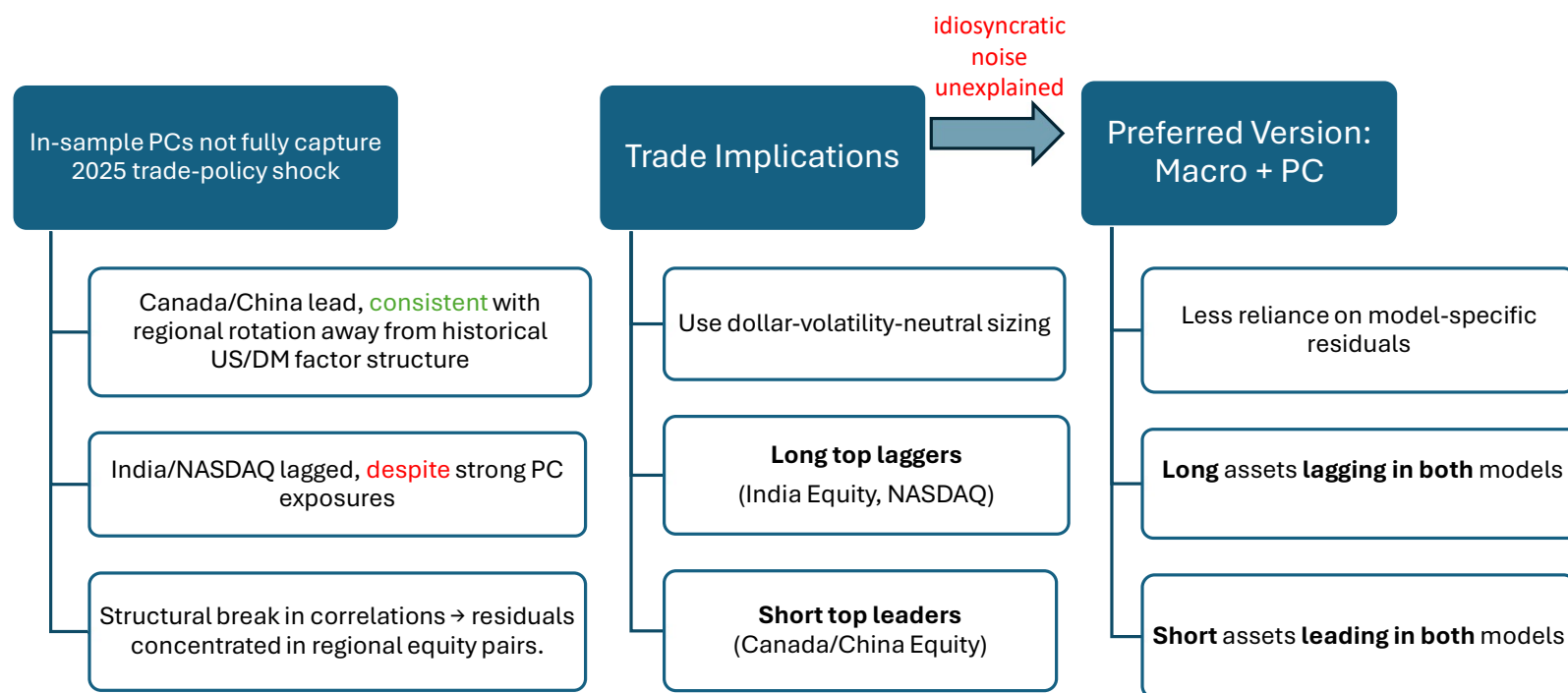
	cum_actual_std	cum_PCpred_std	cum_resid_std	in_sample_R2
Canada_Eq	8.41	6.41	2.00	0.77
China_Eq	3.90	2.11	1.79	0.39
Brazil_Eq	5.85	4.79	1.06	0.58
US_10Y_TR	3.66	2.92	0.74	0.75
UK_Eq	6.35	5.80	0.55	0.76
GlobalCommod	1.88	1.77	0.11	0.65
Japan_Nikkei	5.84	6.63	-0.79	0.77
EuroArea_Eq	5.13	6.33	-1.21	0.74
USD_Broad	-5.58	-4.25	-1.33	0.58
WTI_Oil	-1.03	2.63	-3.66	0.55
US_NASDAQ	3.96	9.82	-5.86	0.77
India_Eq	1.05	7.51	-6.46	0.57

Table: Cumulative OOS Performance vs. PC-Factor Prediction

- Median in-sample $R^2 = 0.69$, much better than macro-factors (0.06-0.64)
- OOS residuals smaller, more concentrated

PC Price Factors

Interpretations & Trade Implications



Conclusion

- Macro model explains a modest but economically intuitive cross-asset returns, while PCA captures co-movement more efficiently; both identify same 2025- pattern
- Residual-based trades should be small tactical overlays, not high-conviction positions, since 2025- dispersion not captured by either model